

Development of the Face

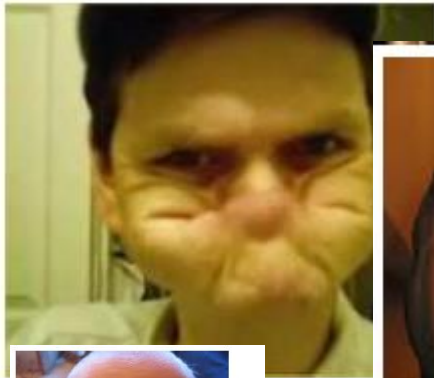
Dr. Nándor Nagy

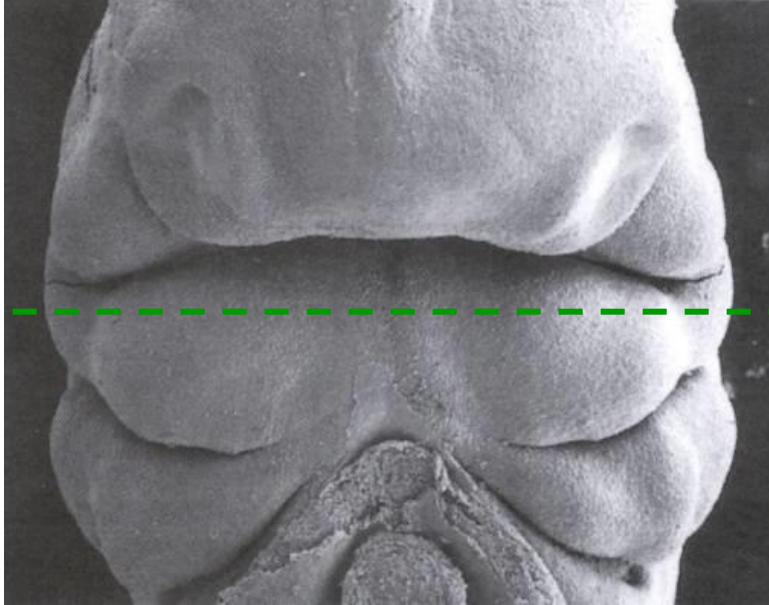


Semmelweis University

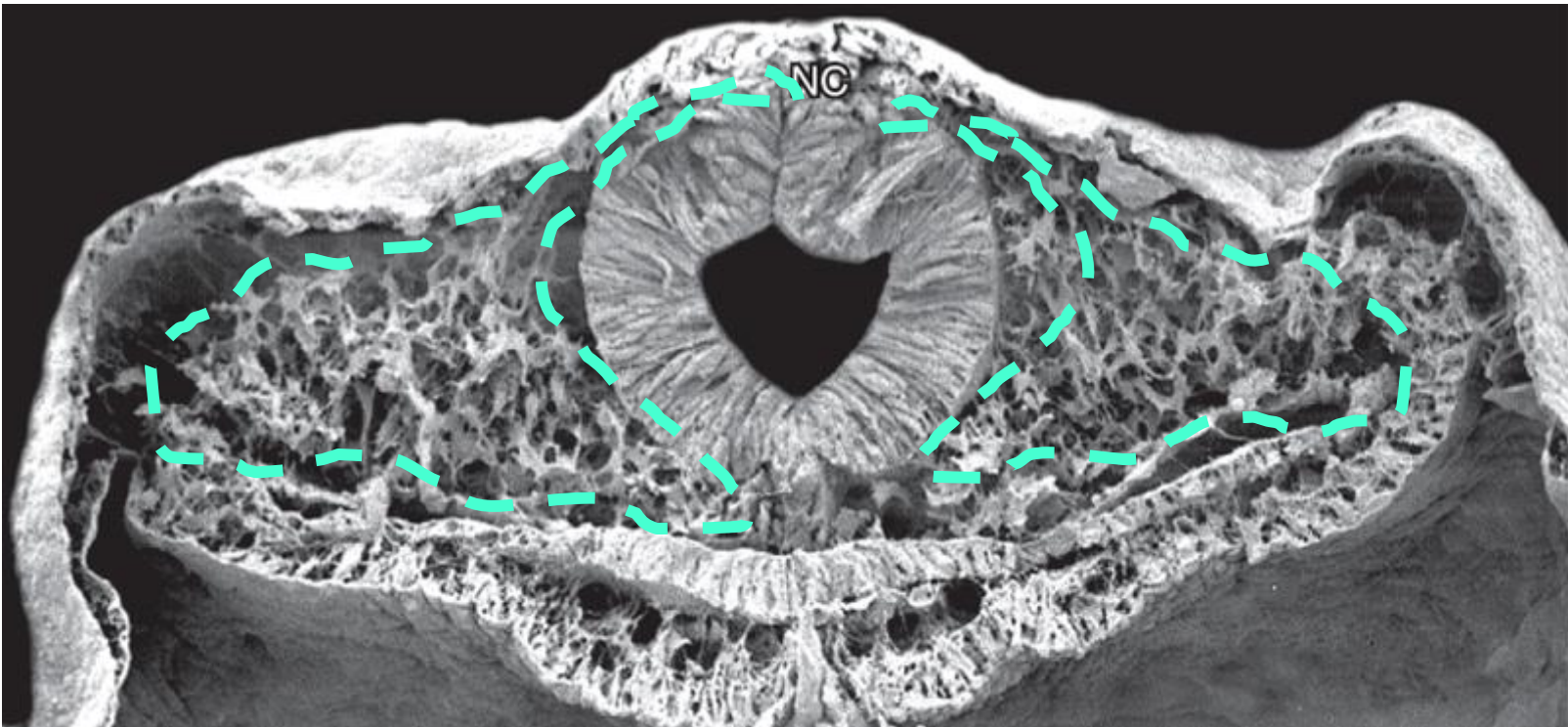


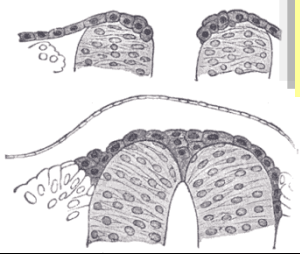
The basic morphology of the face is established between the 4th and 10th weeks by the development and fusion of **five prominences (processes)**.





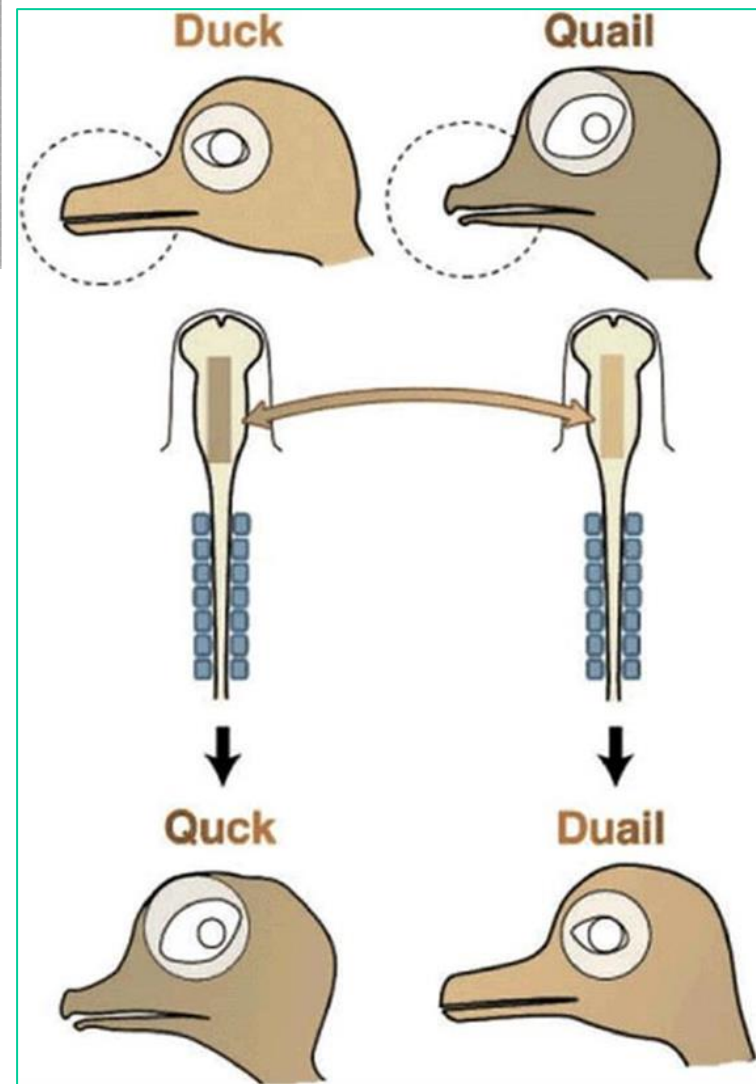
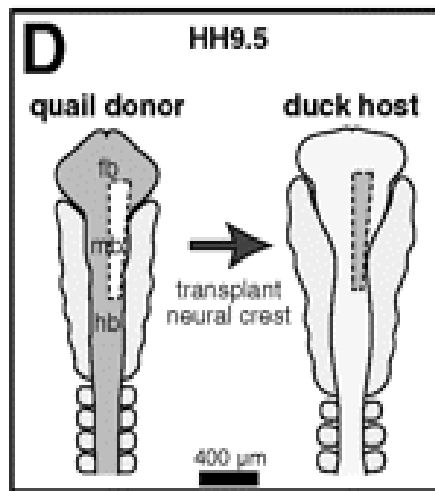
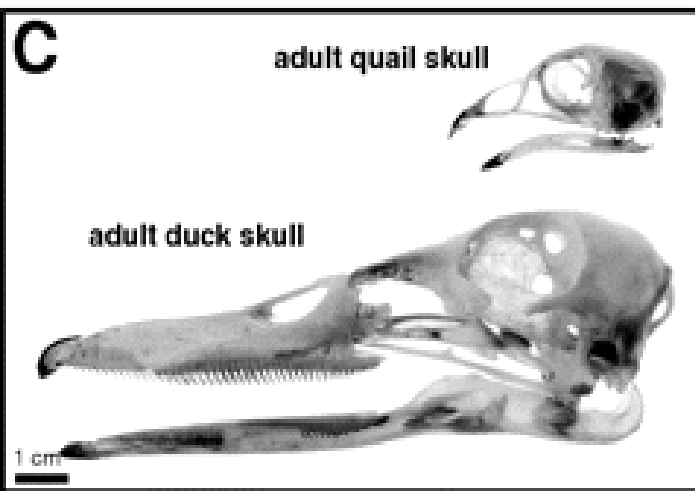
Neural crest: the largest contributor to the developing face, gives rise to cartilage, bone, dentin and pulp of the teeth, dermal papillae of hair follicles, smooth muscles of the vessel walls, ligaments, fascia, adipose tissue, dermis, pericytes in the forebrain, pigment cells, peripheral glial cells, subpopulation of sensory neurons

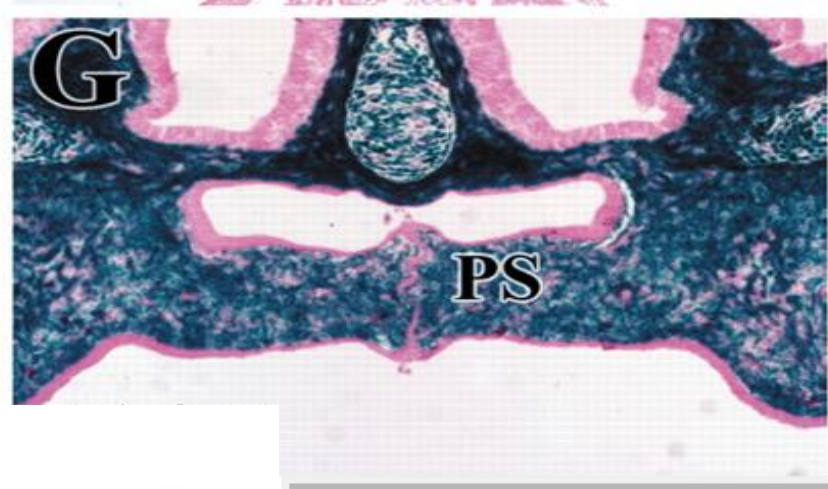
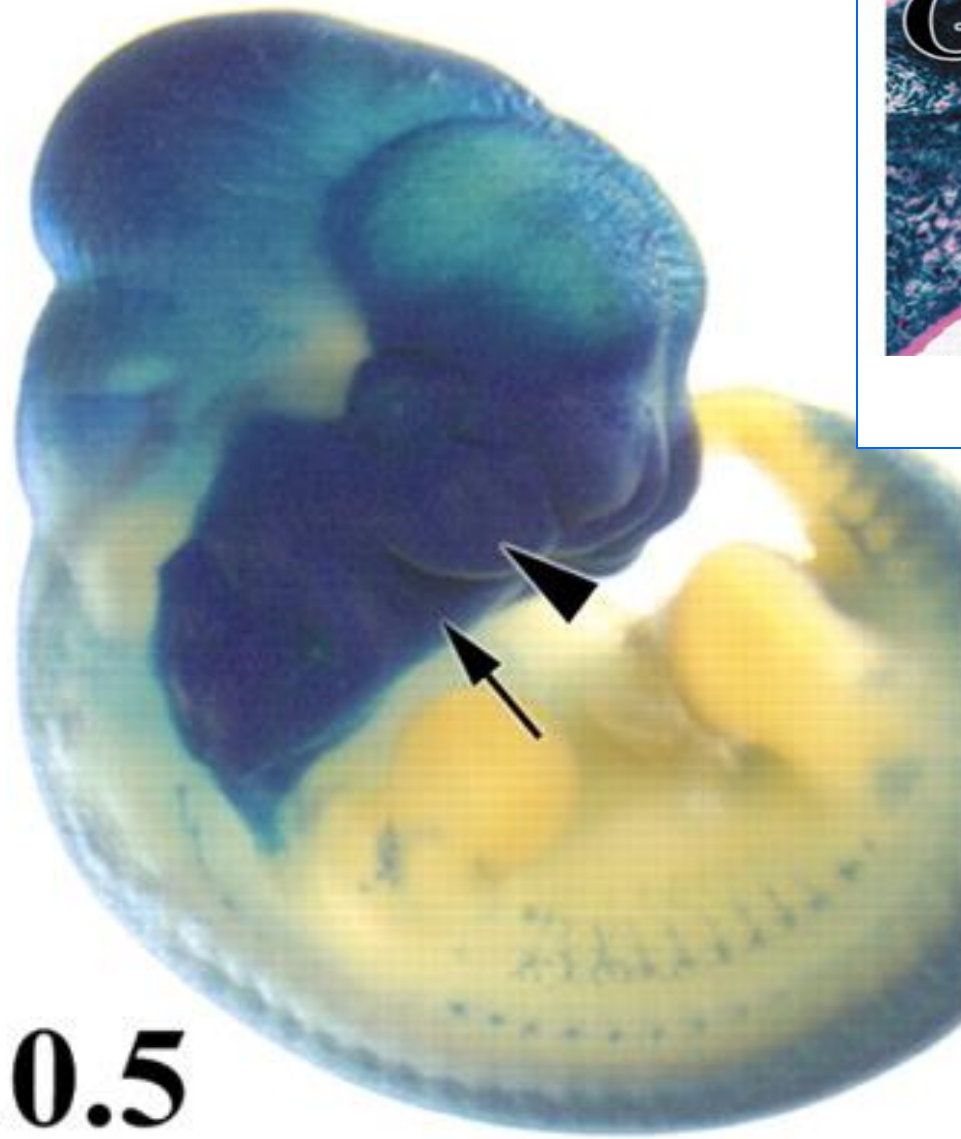




1. Migration of cranial neural crest



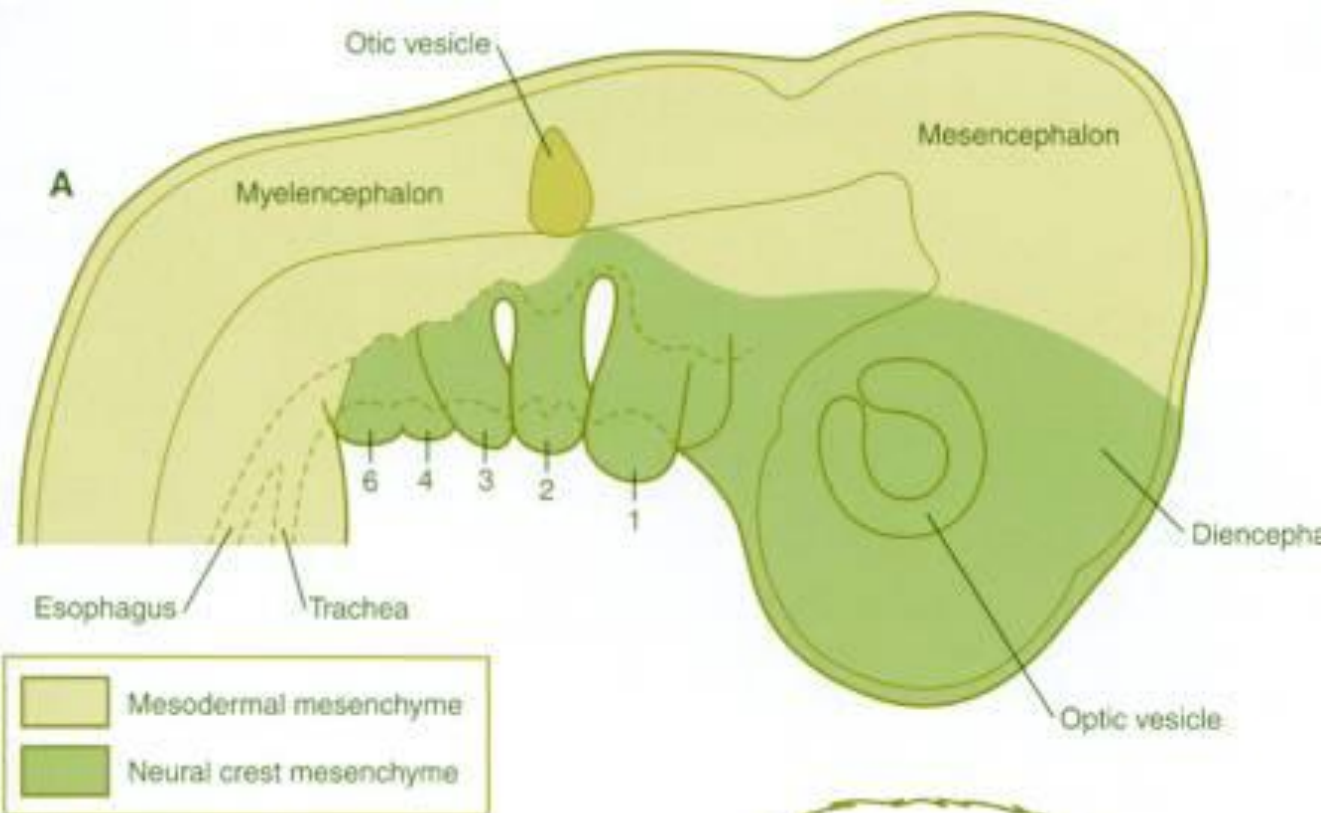




**Distribution of
Wnt1 expressing
*cranial neural
crest cells* in
developing
mouse embryo**

E10.5

Fate mapping of NCCs using Wnt1 gene expression shows that facial bones and cartilage, and frontal bones, stain blue and are derived from the neural crest.



A single gene controls how our faces develop when we are young



HUMANS 4 December 2019

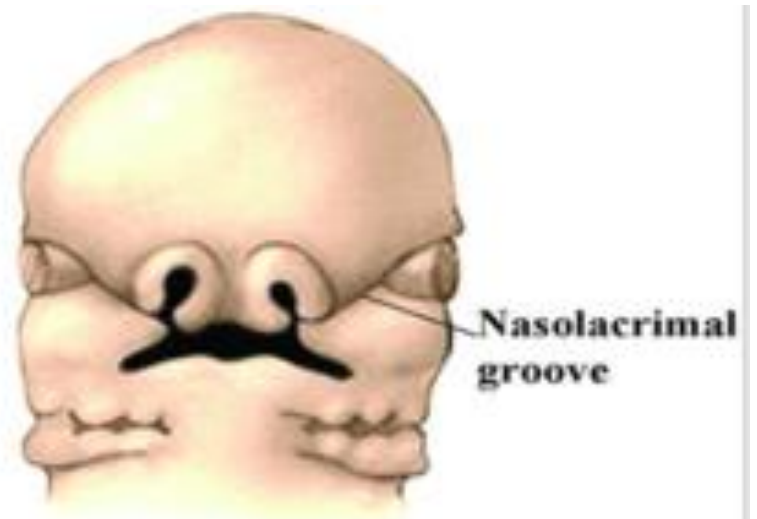
By [Michael Marshall](#)



SCIENCE ADVANCES | RESEARCH ARTICLE

EVOLUTIONARY BIOLOGY

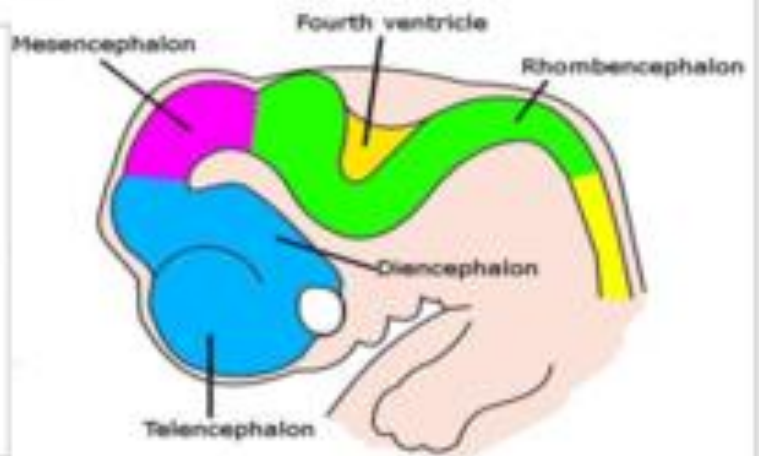
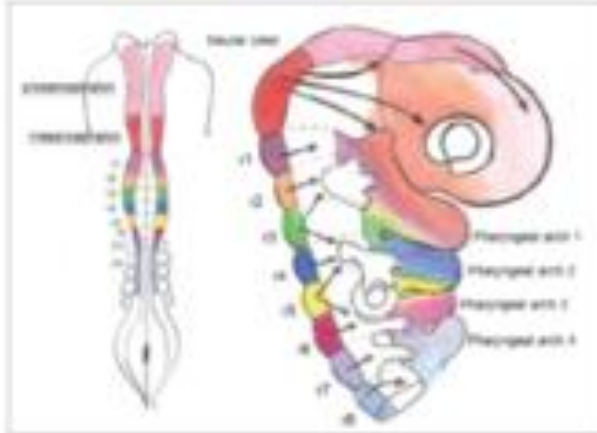
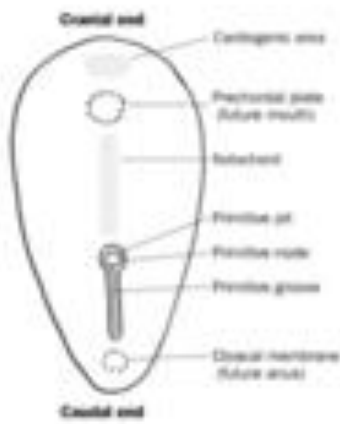
Dosage analysis of the 7q11.23 Williams region identifies *BAZ1B* as a major human gene patterning the modern human face and underlying self-domestication



A

B

C



4th week

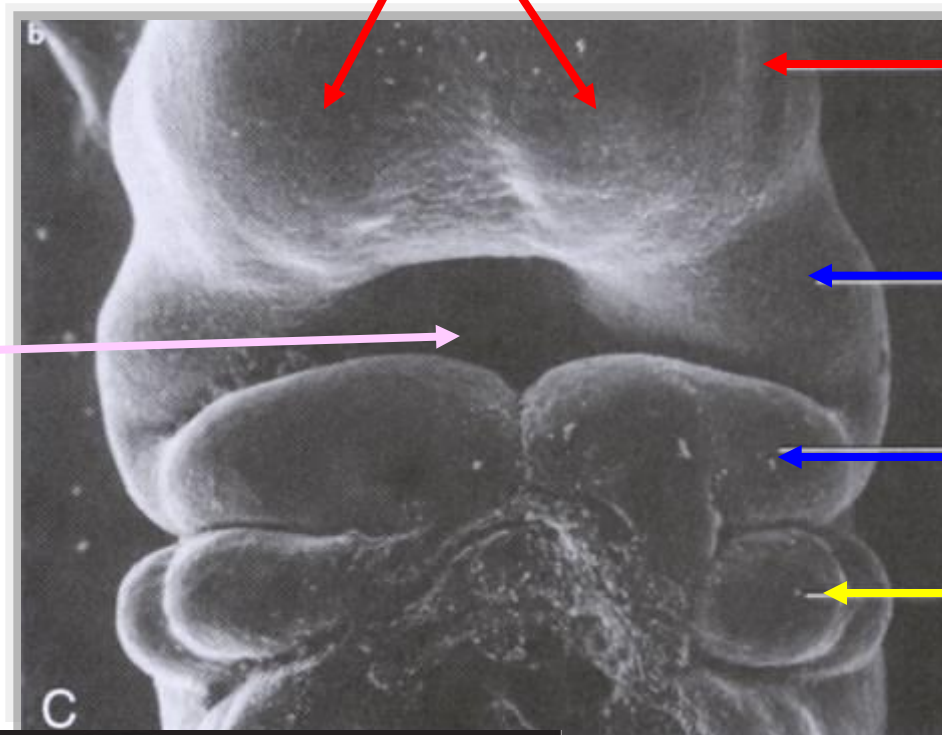
-processus frontalis (frontonasal prominence)

-stomodeum
(primitive oral cavity)

Nasal placod
(Pax-6)

-processus maxillaris
(maxillary
Prominences)

-processus mandibularis
(two mandibular prominence)



Frontonasal
prominence

Nasal pit

Disintegrating
oropharyngeal
membrane

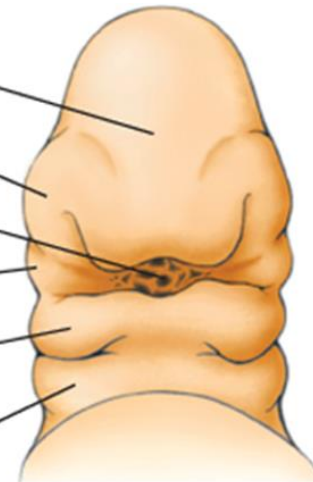
Maxillary
prominence

Mandibular
prominence

2nd pharyngeal
arch

C

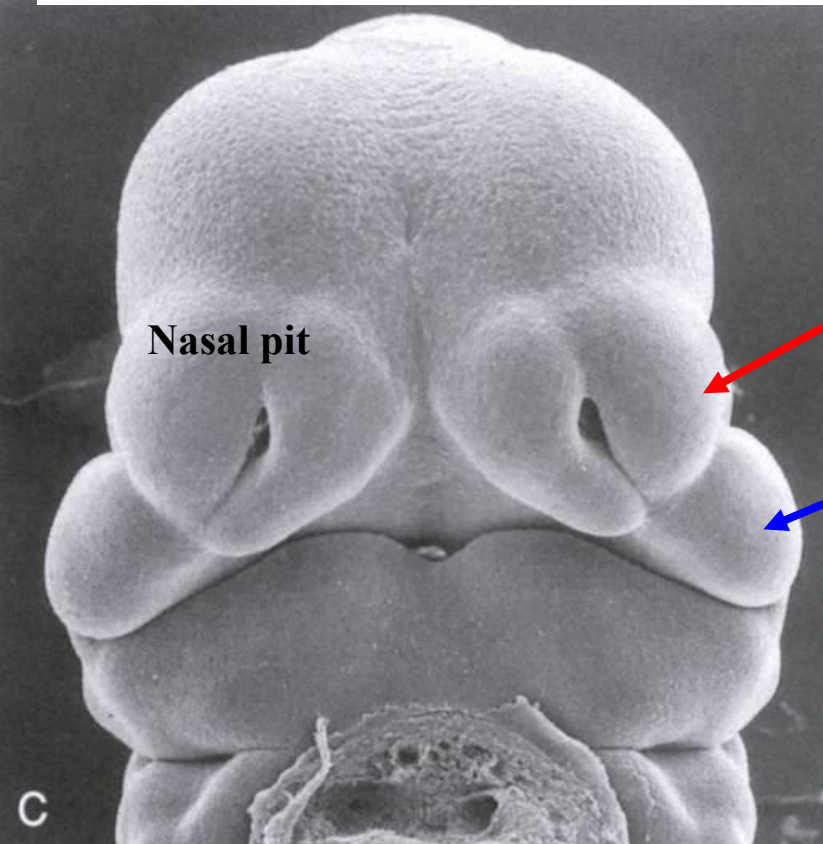
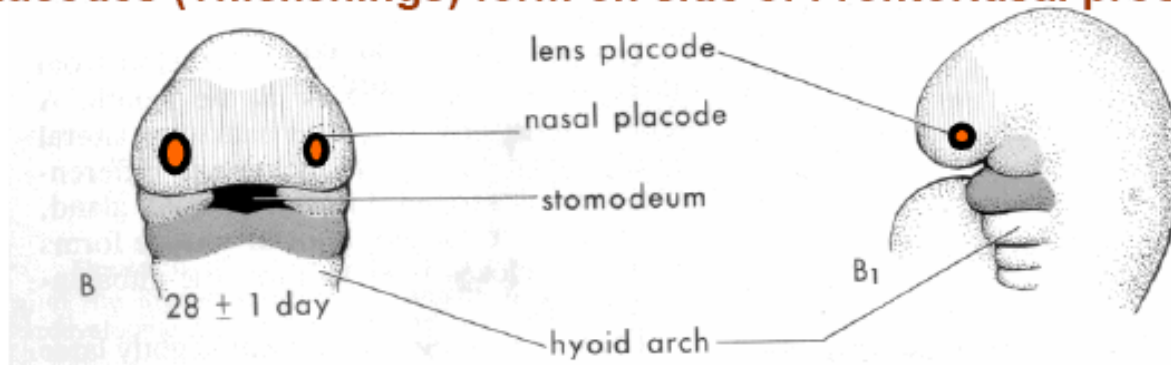
5th week





1. Nasal Placodes (Thickenings) form on side of FrontoNasal process

Nasal Placode



Nasal or olfactory placodes invaginate to form the nasal pit, dividing the frontonasal prominence into the lateral and medial nasal process. (Nasal placodes give rise to the olfactory epithelium)

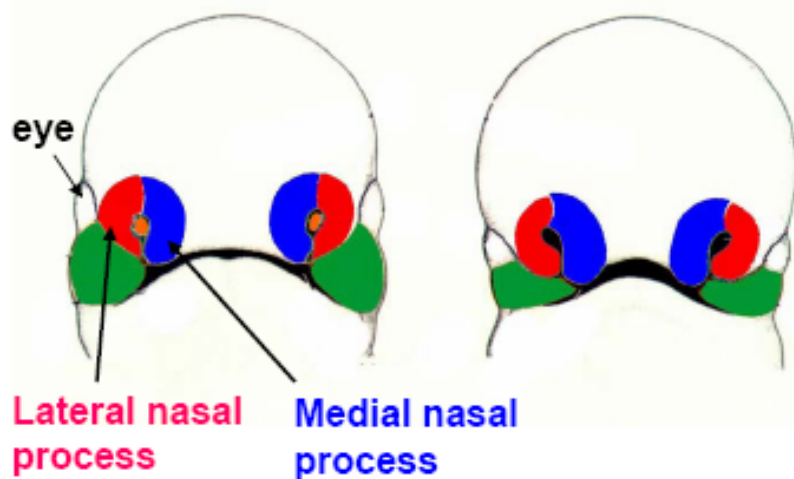
maxillary prominences

-proliferate and move toward the midline

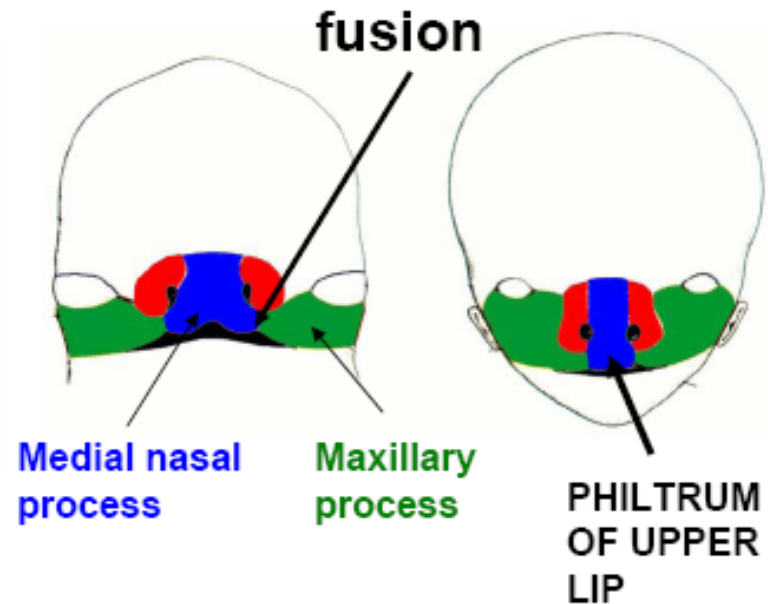
5-6th week

DEVELOPMENT OF FACE

2. Medial & Lateral Nasal Processes—form at margins of nasal placodes

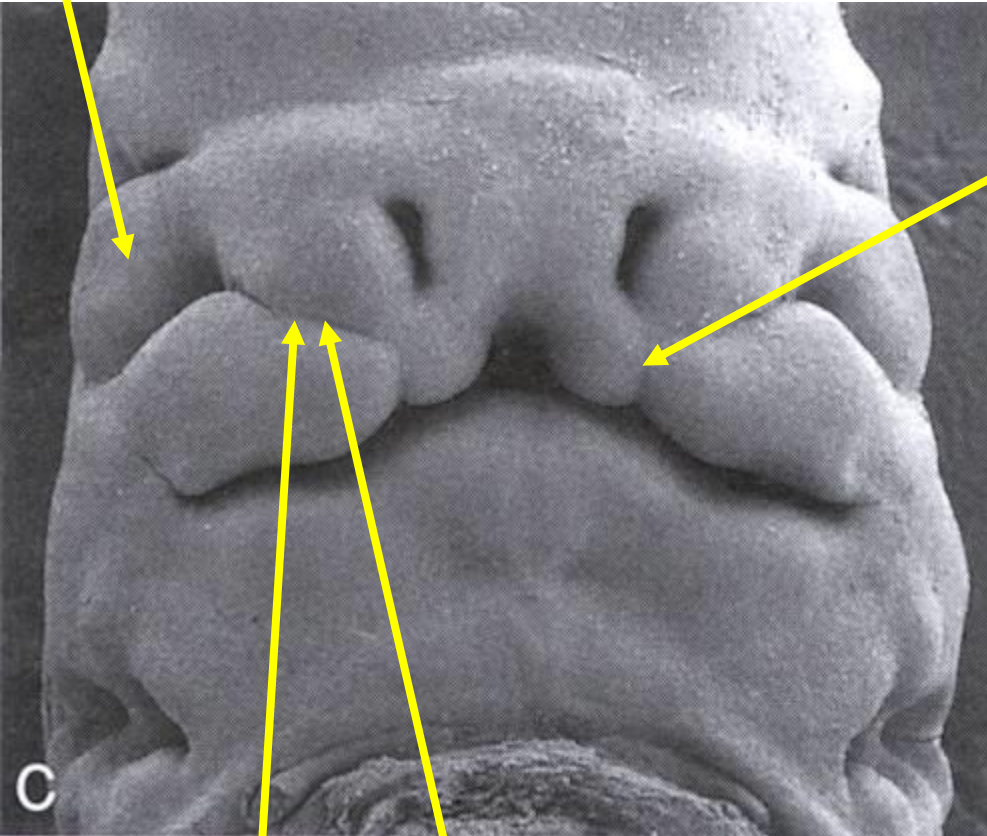


3. Medial nasal process & Maxillary Process—fuse to form upper lip



7th week

eye



Medial nasal process and maxillary process

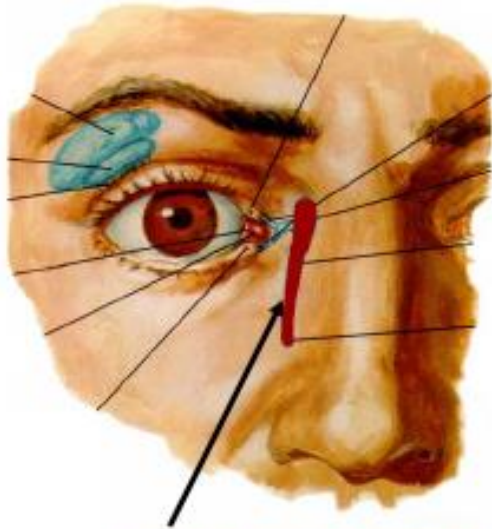
Upper lip: 2 Medial nasal process
2 processus maxillaris

Lower lip: fusion of 2 mandibular process

- Nasolacrimal groove:
the ectoderm invaginates and form a duct, that is
invested by bone during the ossification of the maxilla

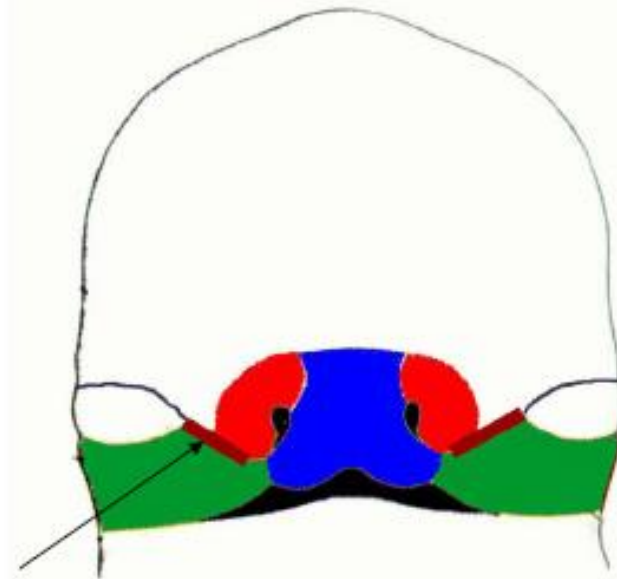
-fusion of the mandibular and lateral nasal process

DEVELOPMENT OF NASOLACRIMAL DUCT



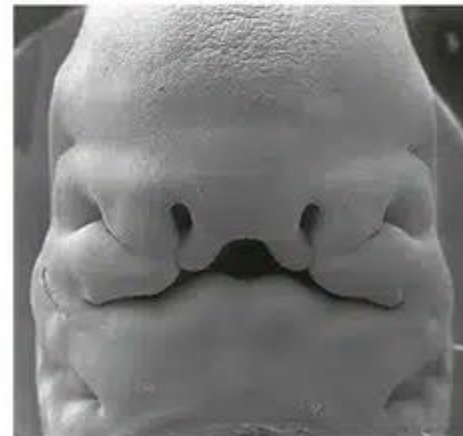
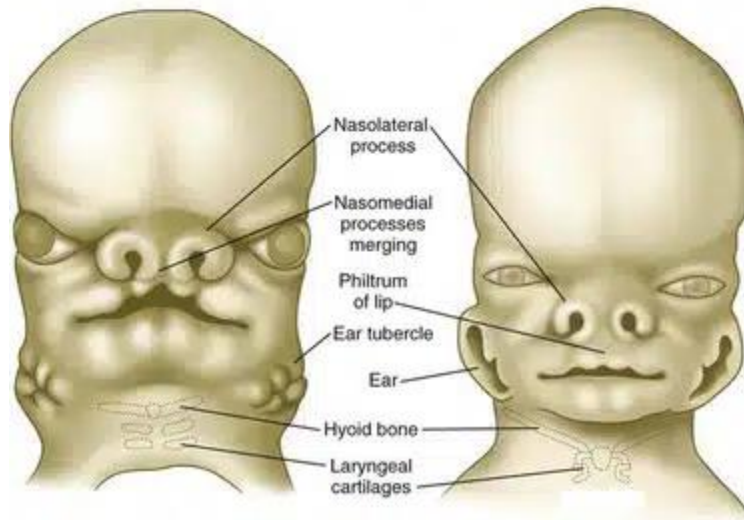
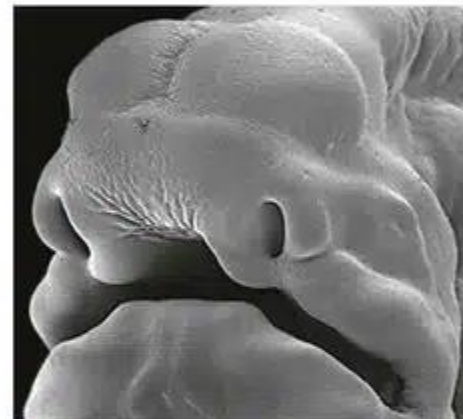
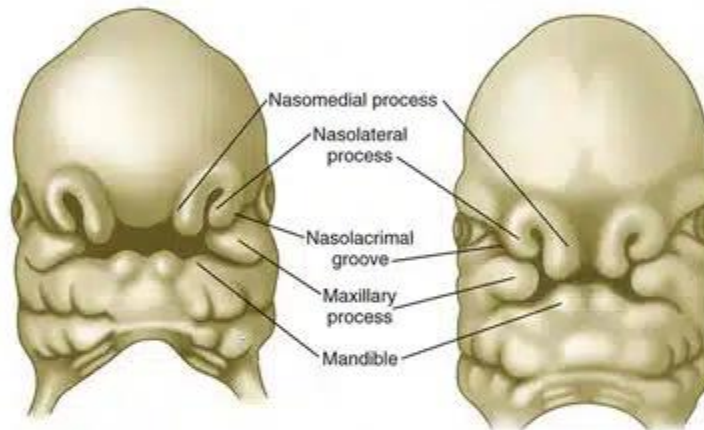
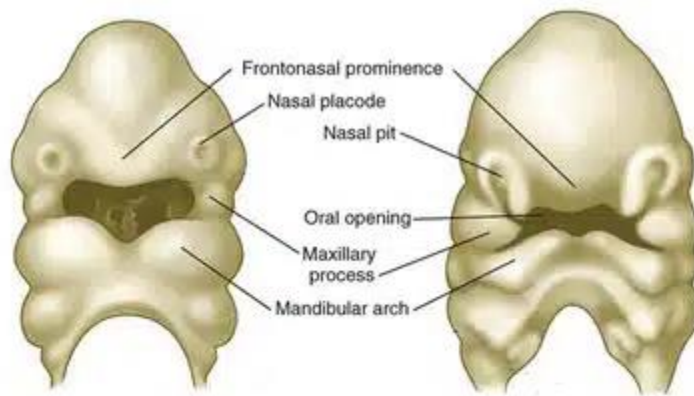
NASOLACRIMAL DUCT

– connects anterior eye to nasal cavity

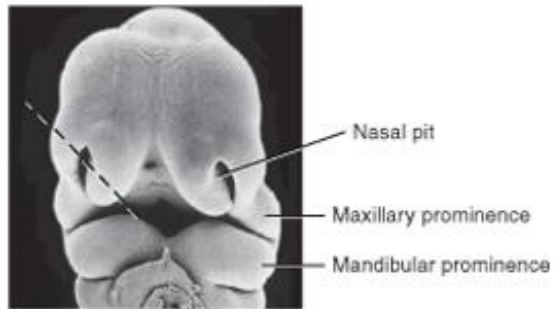


- Develops as solid cord from medial angle of eye to nasal cavity
- becomes canalized.

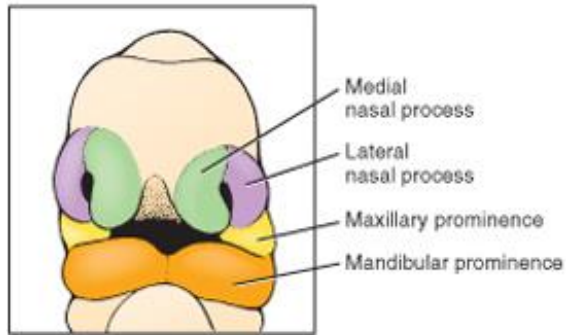
**Obstructed Duct - failure of duct to canalize;
opened surgically for tears to drain to nasal cavity**



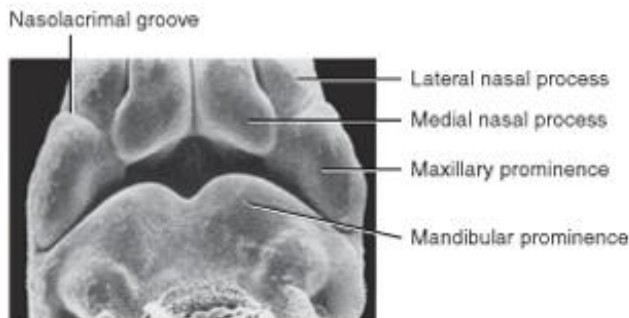
Prominence	Derivatives
Frontonasal	Forehead, bridge of nose, medial and lateral nasal prominences
Medial nasal	Philtrum, primary palate, upper 4 incisors and associated jaw
Lateral nasal	Sides of the nose
Maxillary (1st pharyngeal arch)	Cheeks, lateral upper lip, secondary palate, lateral upper jaw
Mandibular (1st pharyngeal arch)	Lower lip and jaw



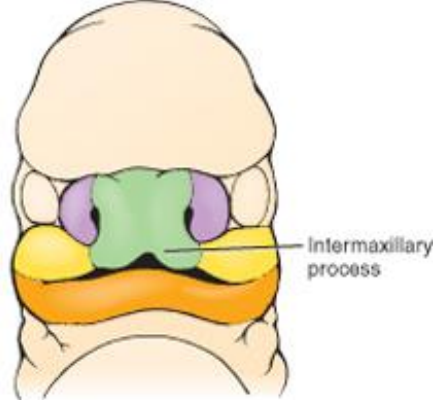
A Early 6th week



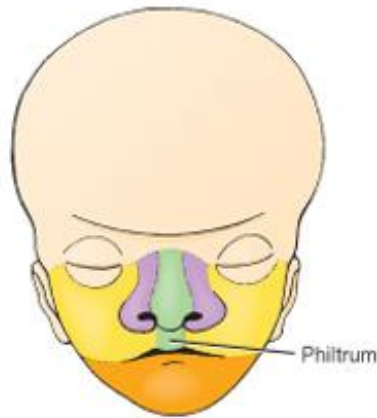
B Early 6th week



C Early 7th week

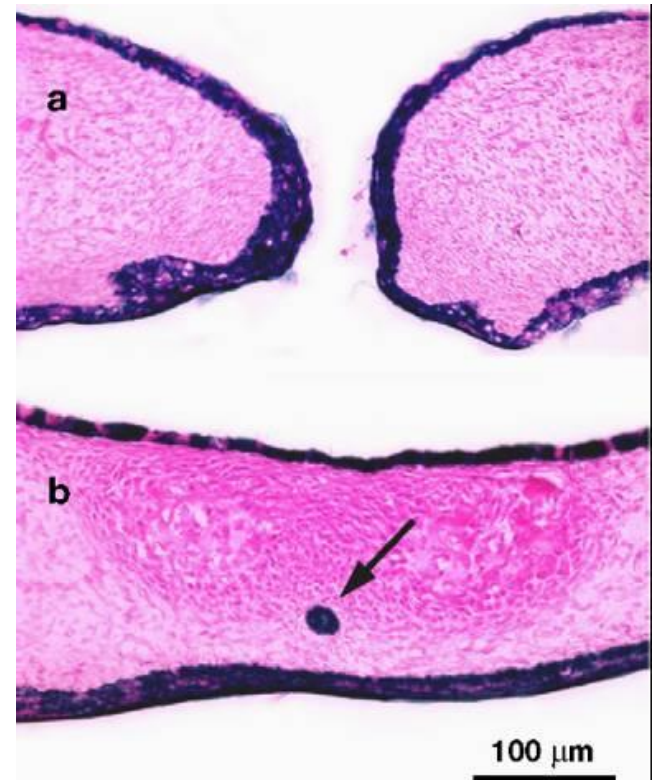


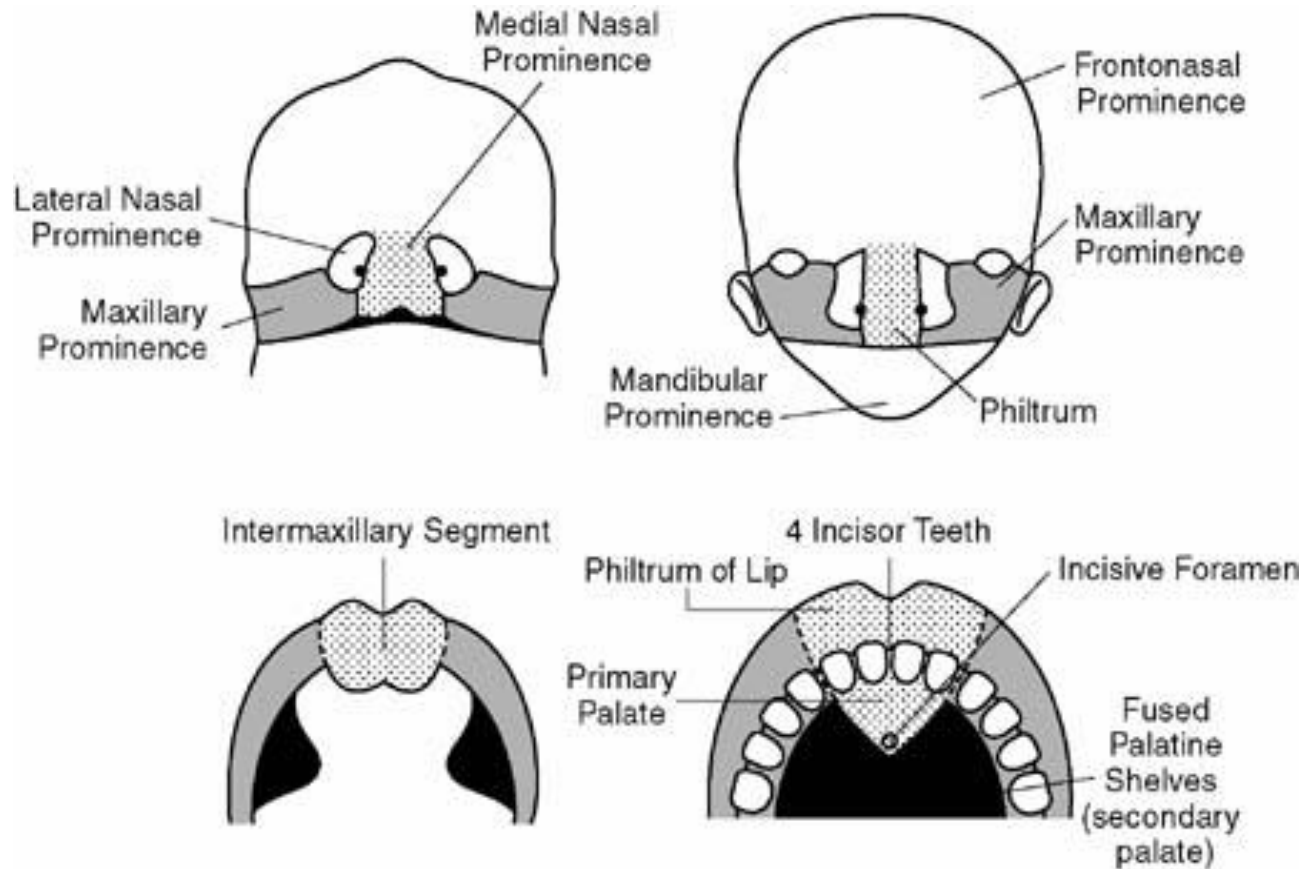
D Late 7th week



E 10th week

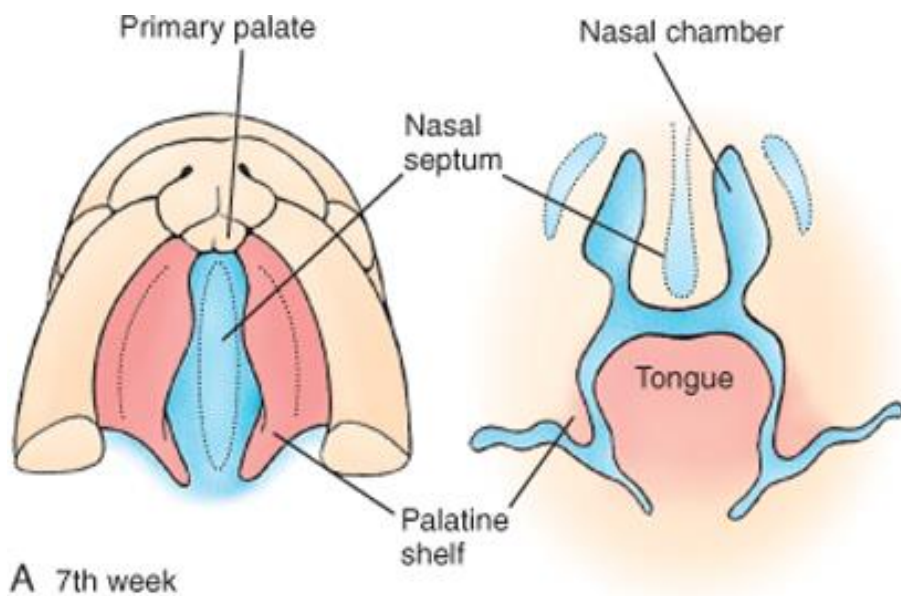
The medial nasal processes fuse to form the septum of the nose and the **intermaxillary process**. The intermaxillary process give rise to the philtrum and the primary palate containing four incisor teeth.



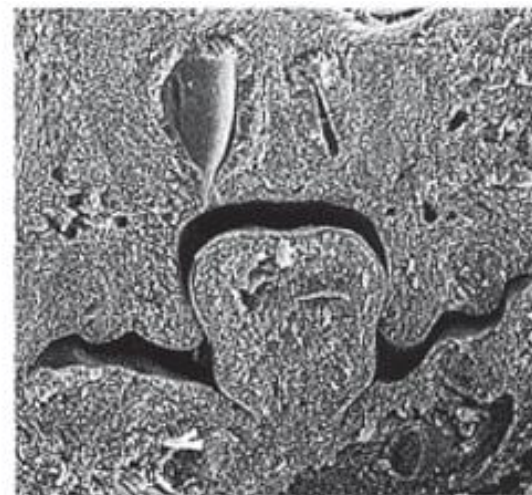


. Palate and Face Development

the **primary palate** will form the premaxillary portion of the maxilla (anterior one-third of the final palate). This small portion is anterior to the incisive foramen and will contain the 4 maxillary incisors.

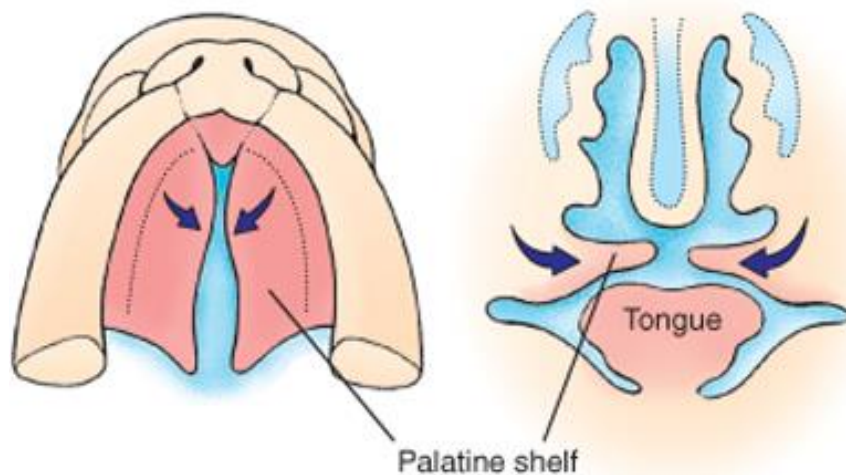


A 7th week

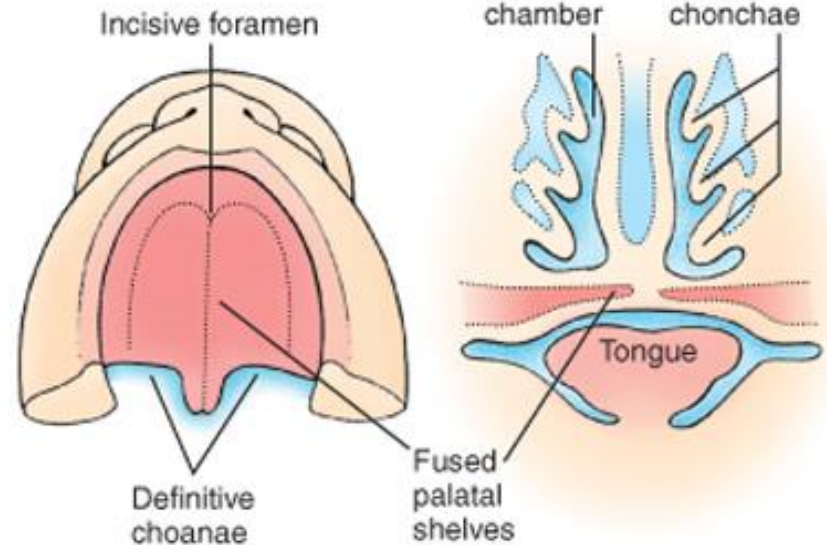


B 7th week

Medial wall of the maxillary prominence form the palatine shelf



C Late 7th/8th week



D 10th week

Clinical Relevance: Cleft Lip and Cleft Palate

A cleft refers to a gap/split in the upper lip or palate. It results from a defect during development of face and palate:

- **Cleft lip** – occurs when the medial nasal prominence and maxillary prominence fail to fuse.
- **Cleft palate** – can occur in isolation when the palatal shelves fail to fuse in the midline, or in combination with cleft lip.

Cleft lip and cleft palate are relatively common, occurring in approximately **1/1000 births**.

In addition to the cosmetic and psychosocial implications, severe cleft lip/palate can be a cause of death if a baby is unable to feed.

Clefts of the lip/palate are common birth defects (1:1000) requiring extensive treatment. They appear to be caused by multiple genetic and environmental factors during face development. This may result in local changes in **growth factors**, extracellular matrix (ECM), and cell adhesion molecules.

Table 1. *Biological Factors in Palatogenesis

(A) Secreted Proteins and Their Signaling Pathways

Component	Functions
TGF β s	Mesenchymal proliferation, ECM production, epithelial differentiation
EGFR, PDGF	MMP production
FGFs	Epithelial proliferation, inhibition of apoptosis
SHH, WNTs	Epithelial differentiation, apoptosis

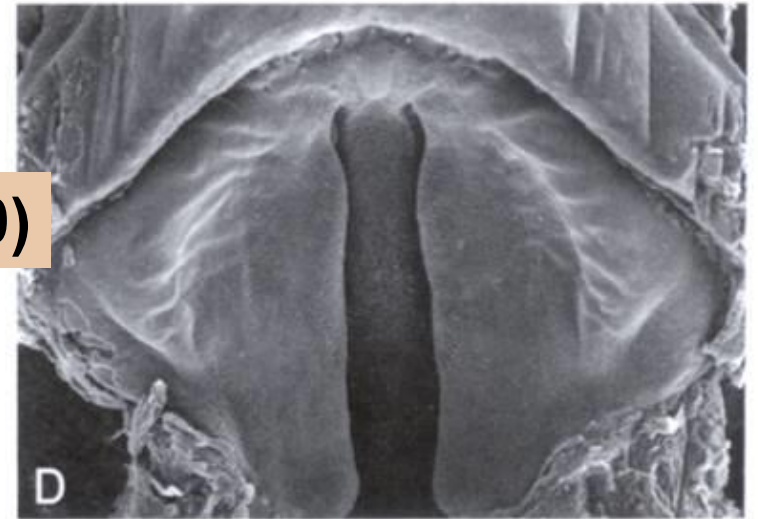
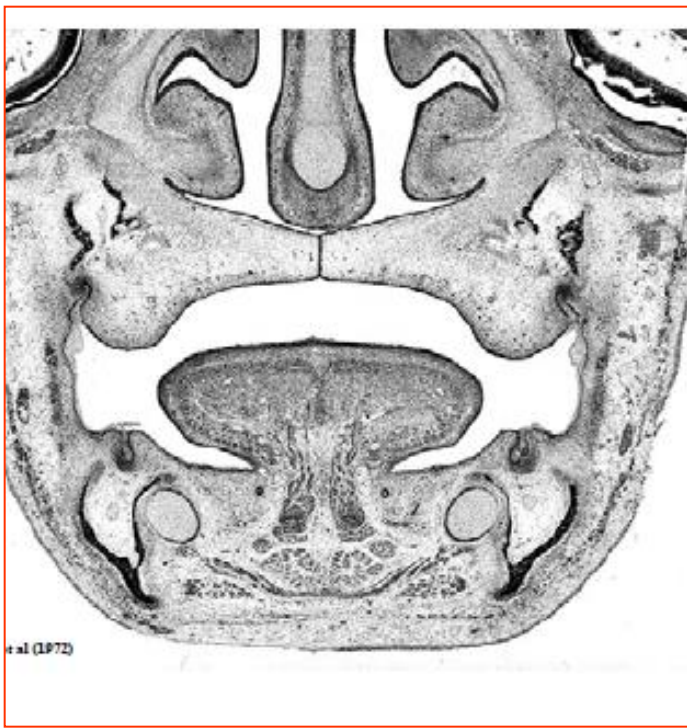
(B) Extracellular Matrix

Component	Functions
Collagens	Mesenchymal growth, shelf elevation
Glycosaminoglycans	Shelf elevation
CSPG	Shelf adhesion

(C) Cell Adhesion Molecules

Component	Functions
Integrins	ECM signaling
E-cadherin	Shelf adhesion
Desmosomal proteins	Shelf adhesion

* This Table summarizes the main biological factors that contribute to palatogenesis, and their biological functions. The factors are

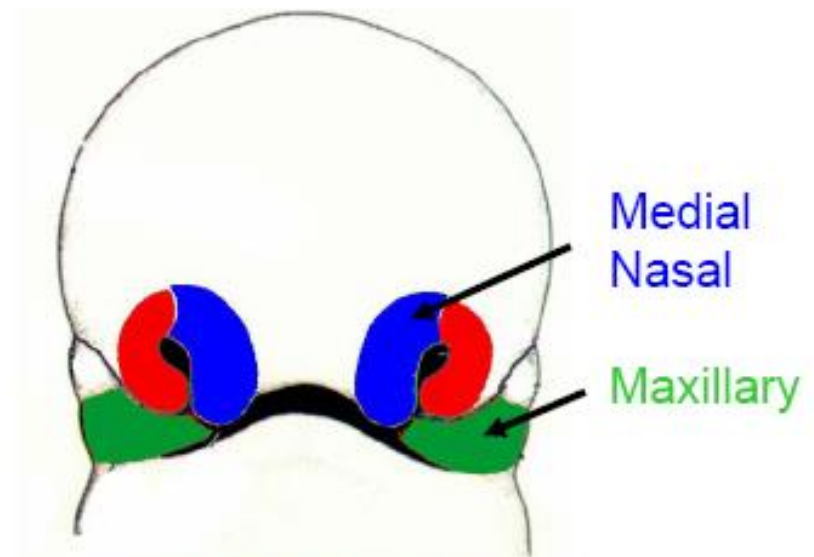


Cleft palate-(1:2500)

The spectrum of malformations ranges in severity from minor midfacial defects to extremely devastating malformations.

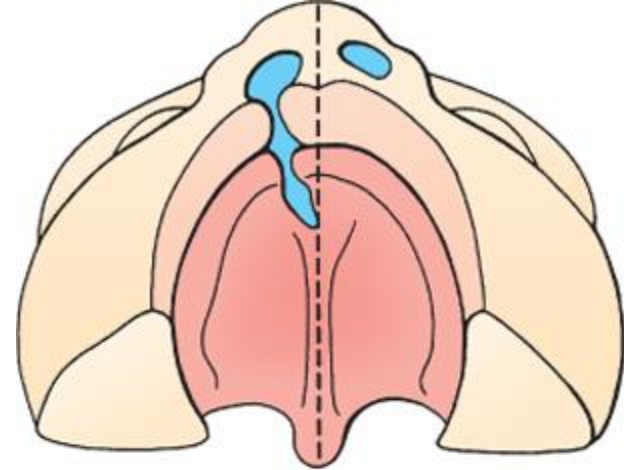
CLEFT LIP

- failure of fusion of Medial Nasal Process and Maxillary process
- 1/1000 Births, can be unilateral or bilateral
- **At** philtrum of lip

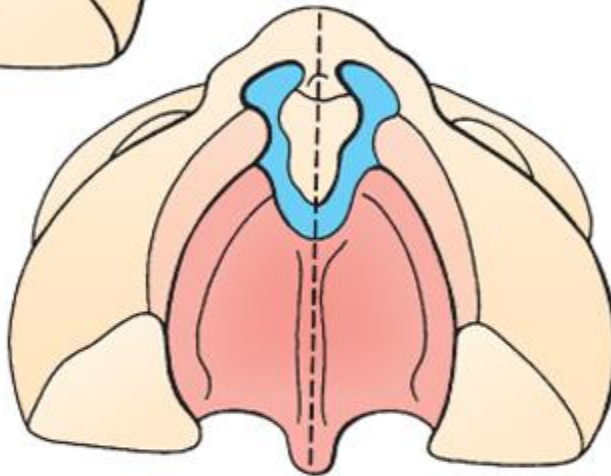


**CLEFT LIP CAN OCCUR
IN COMBINATION WITH
CLEFT PALATE**

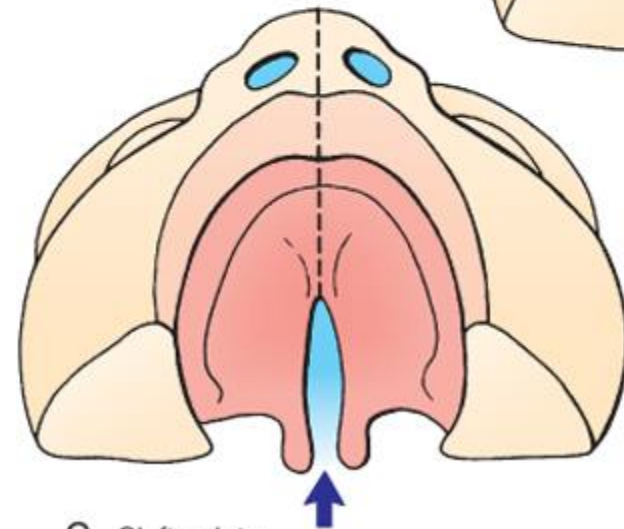




A Unilateral cleft lip

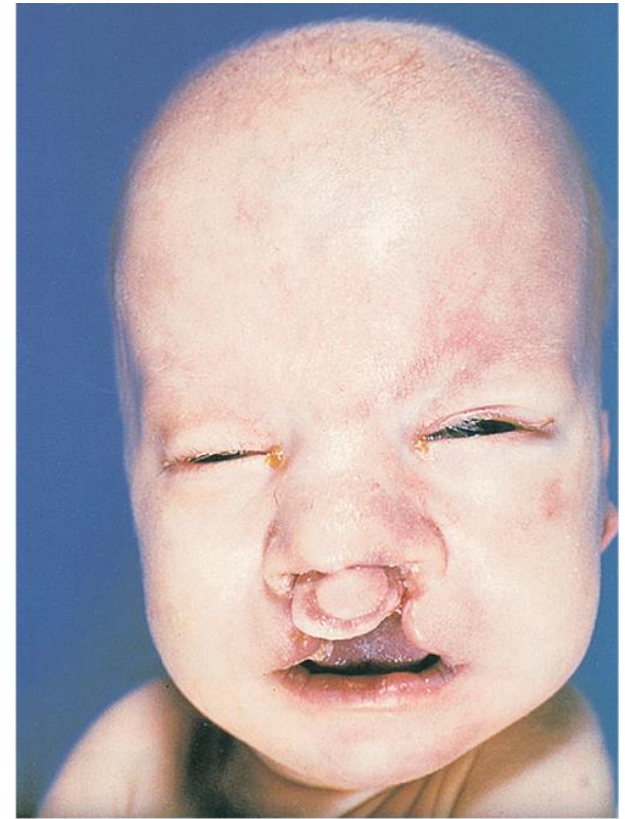


B Bilateral cleft lip
extending to incisive
foramen



C Cleft palate

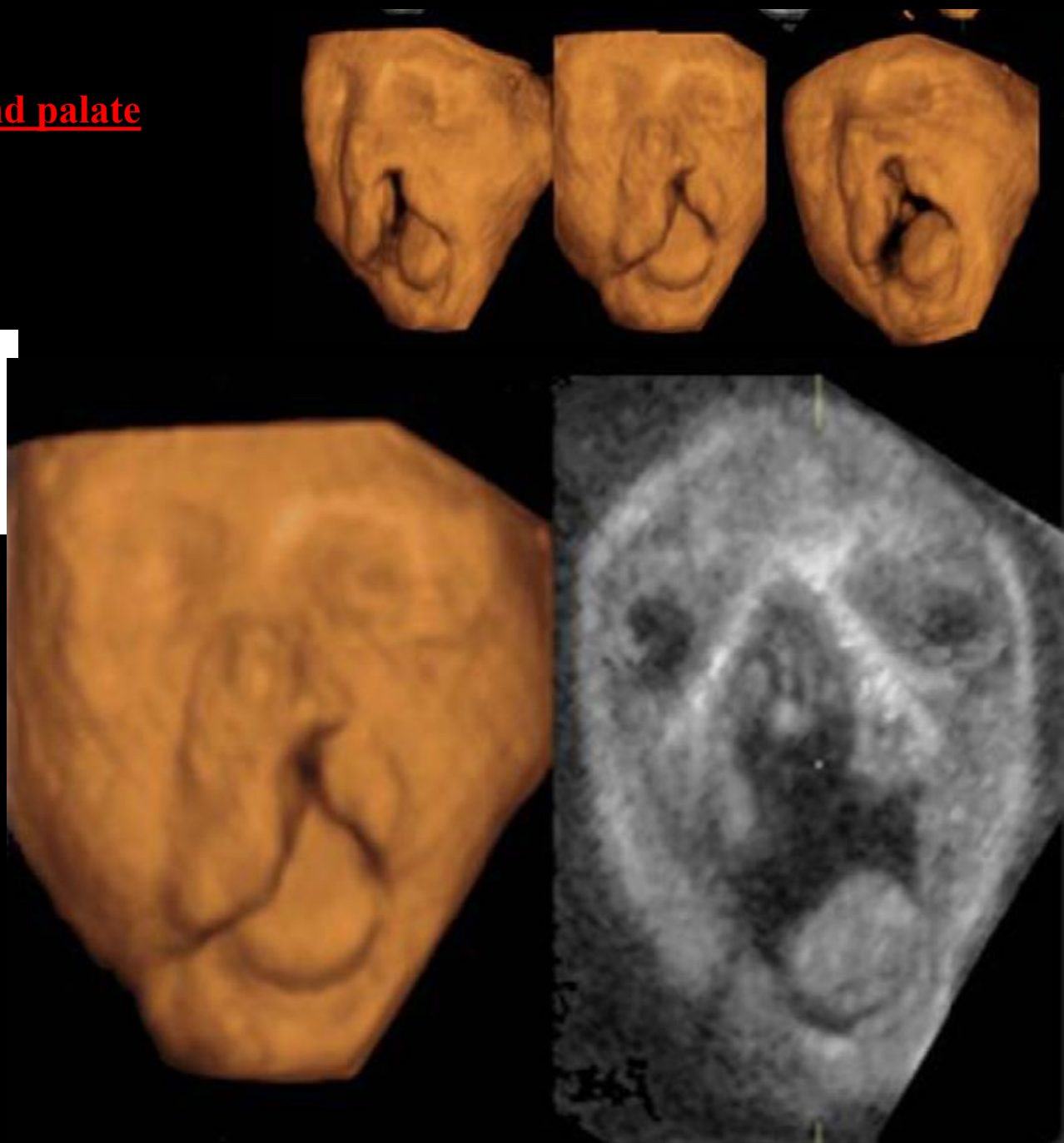
Schoenwolf et al: Larsen's Human Embryology, 4th Edition.
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Bilateral cleft lip. This malformation results from failure of the medial nasal processes to fuse with the maxillary swellings (prominences, processes).

Multiplanar imaging in severe unilateral cleft lip and palate

Three- and four-dimensional
ultrasonography for the structural and
functional evaluation of the fetal face.
Asim Kurjak, et al., Am J Obstet
Gynecol 2007;196:16-28.



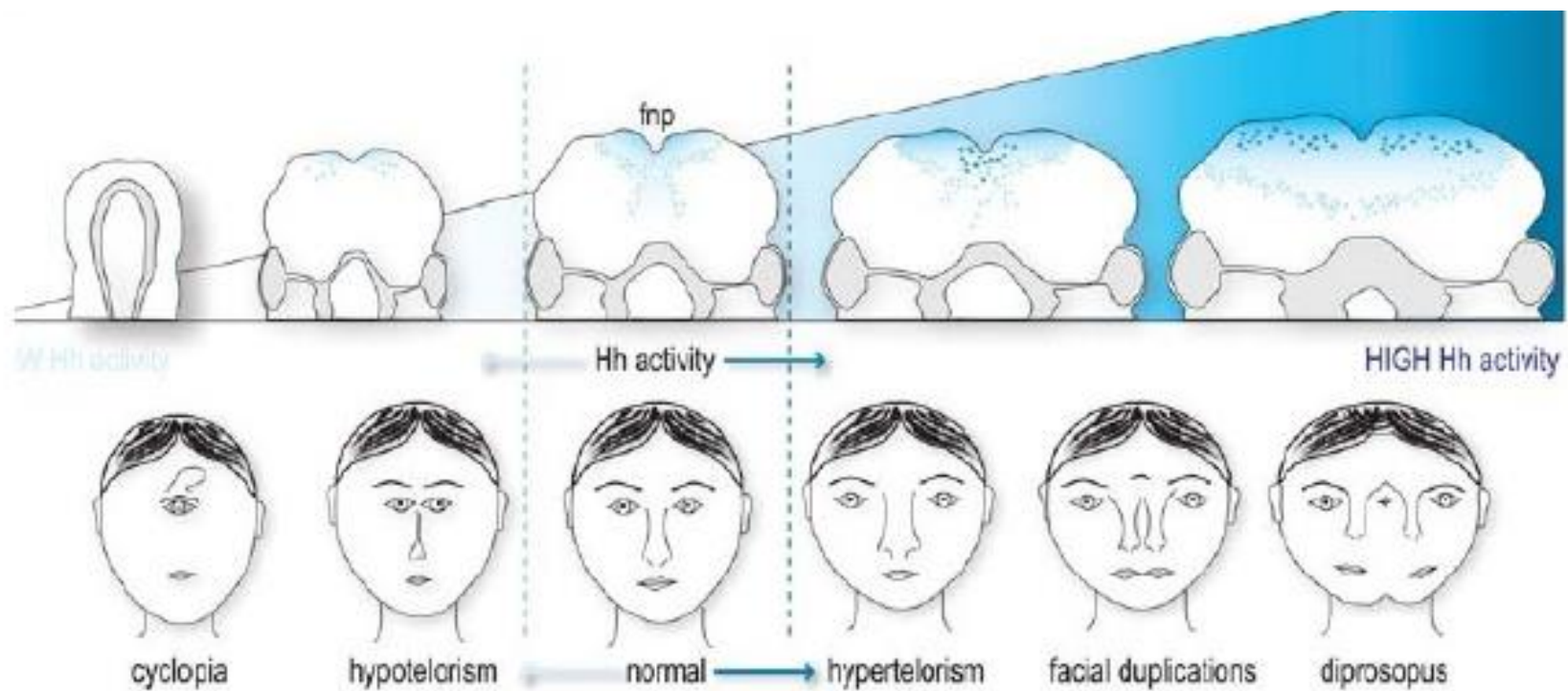
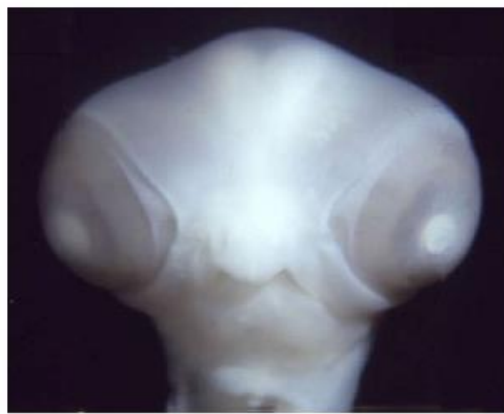


Figure 9 from Brugmann et al., *Human Molecular Genetics* 2010, Vol. 19, No. 8 1577–1592 doi:10.1093/hmg/ddq030 After Wilder



Schoenwolf et al: Larsen's Human Embryology, 4th Edition.
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Journal of
*Developmental
Biology*



Article

Mutation in the Ciliary Protein C2CD3 Reveals Organ-Specific Mechanisms of Hedgehog Signal Transduction in Avian Embryos

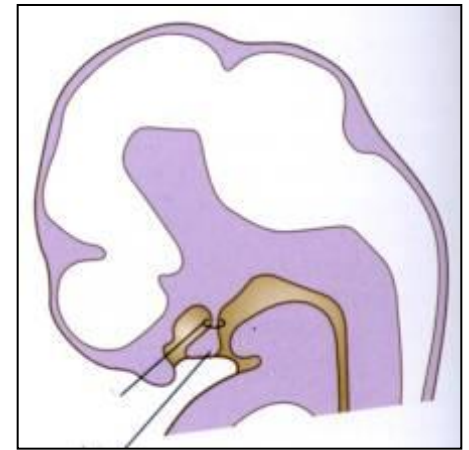
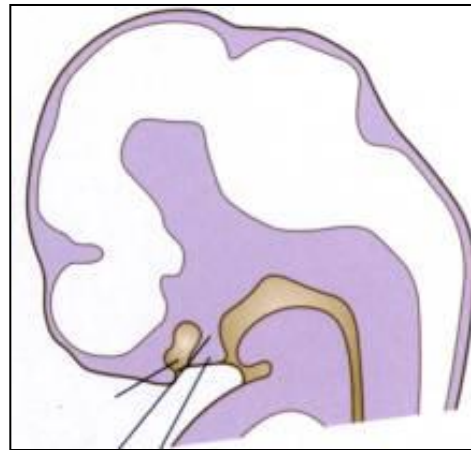
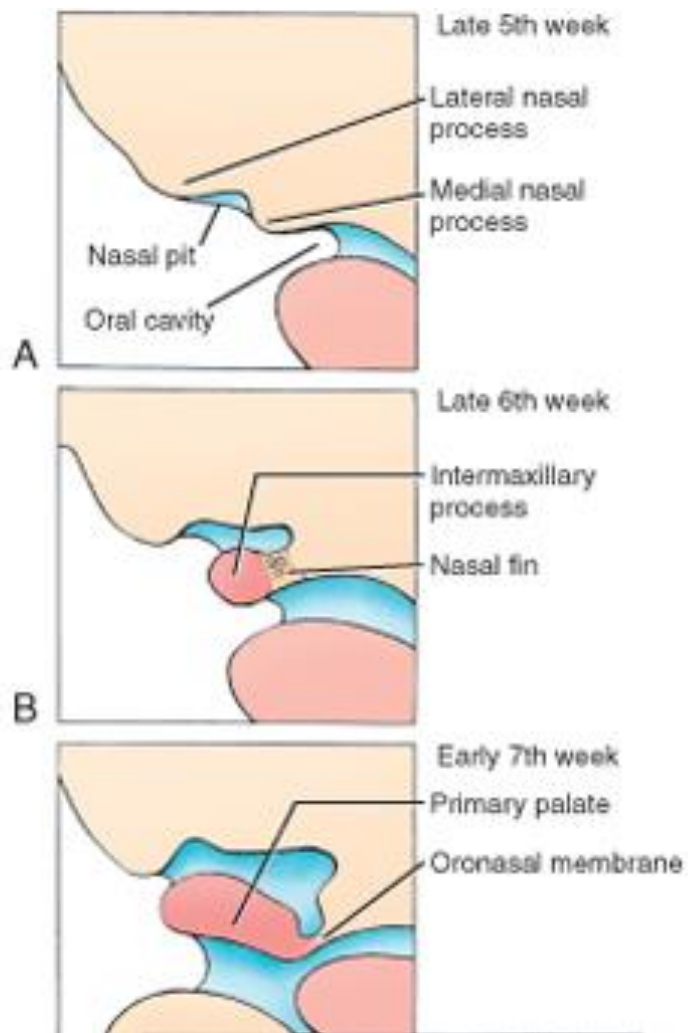
Evan C. Brooks ^{1,2} , Christian Louis Bonatto Paese ^{1,2}, Anne H. Carroll ^{1,2}, Jaime N. Struve ^{1,2}, Nandor Nagy ³ 
and Samantha A. Brugmann ^{1,2,4,*}



holoprosencephaly. Is a birth defect that causes a developing baby's brain to not properly separate into the right and left hemispheres.

—it can also affect the development of an unborn baby's head and face.

1:20.000



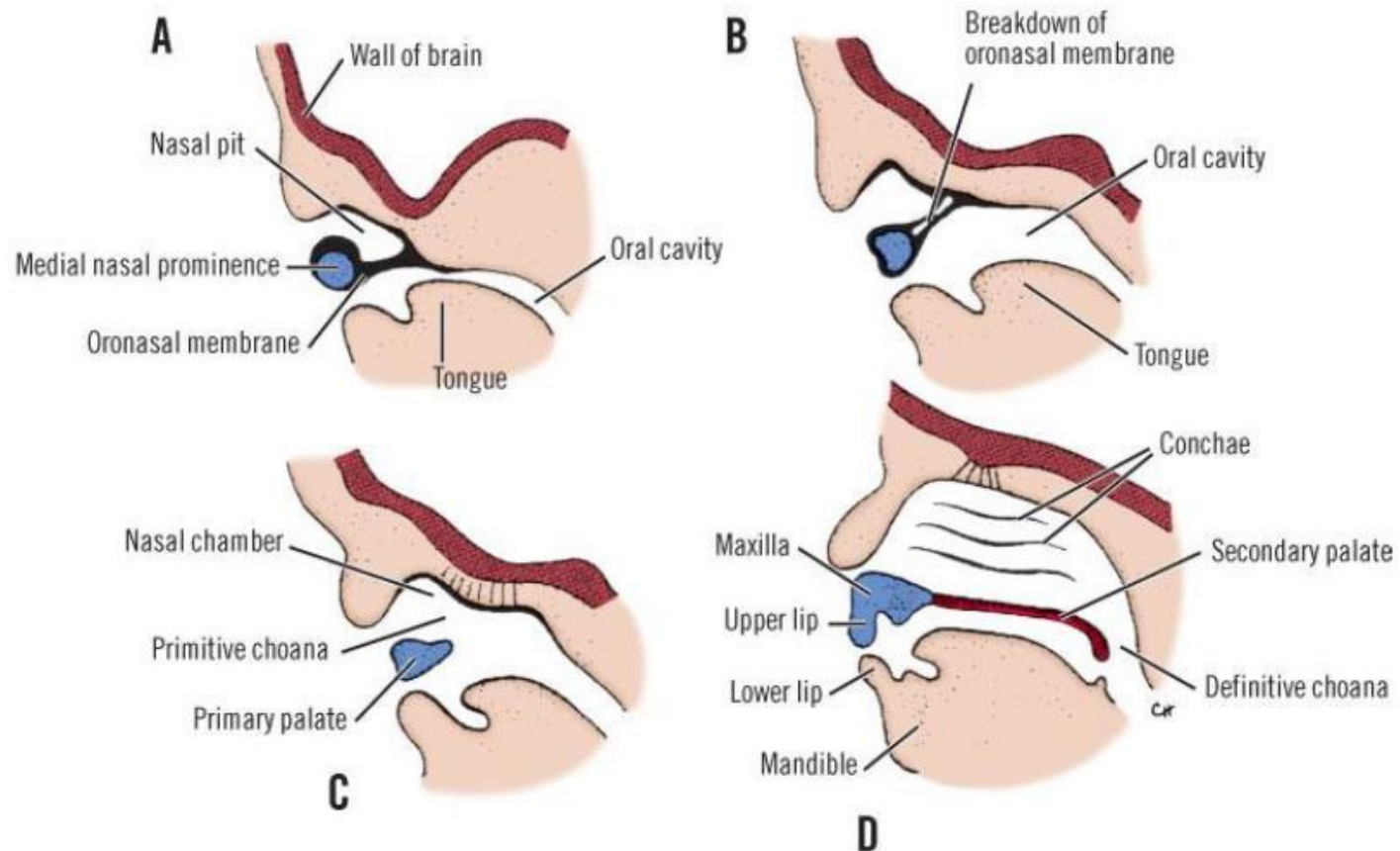
- Nasal sac develops from the nasal pits
- rupture of the oronasal membrane form the primitive choana

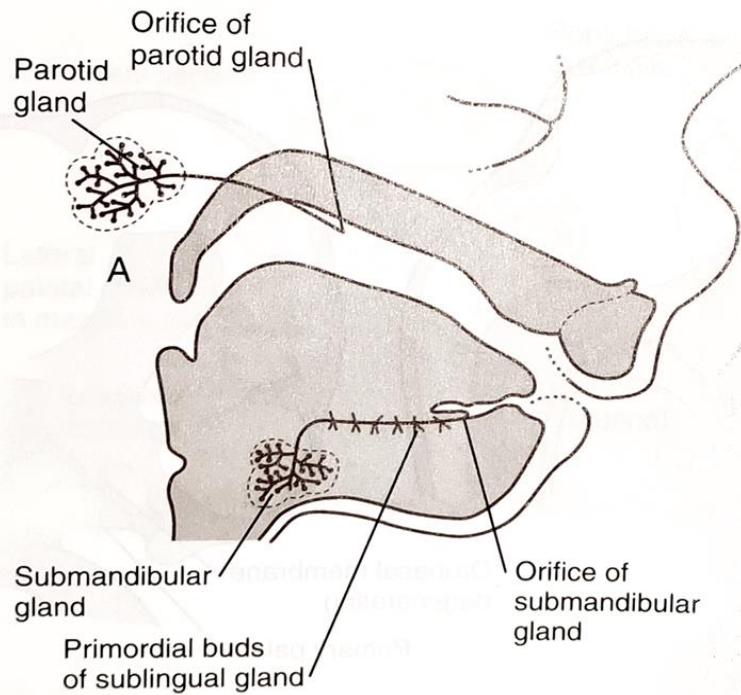
DEVELOPMENT OF THE NASAL CAVITIES AND THE HARD PALATE

By **week 5** the placodes form the **nasal pits**. They further invaginate and the *pits* approach the primitive oral cavity.

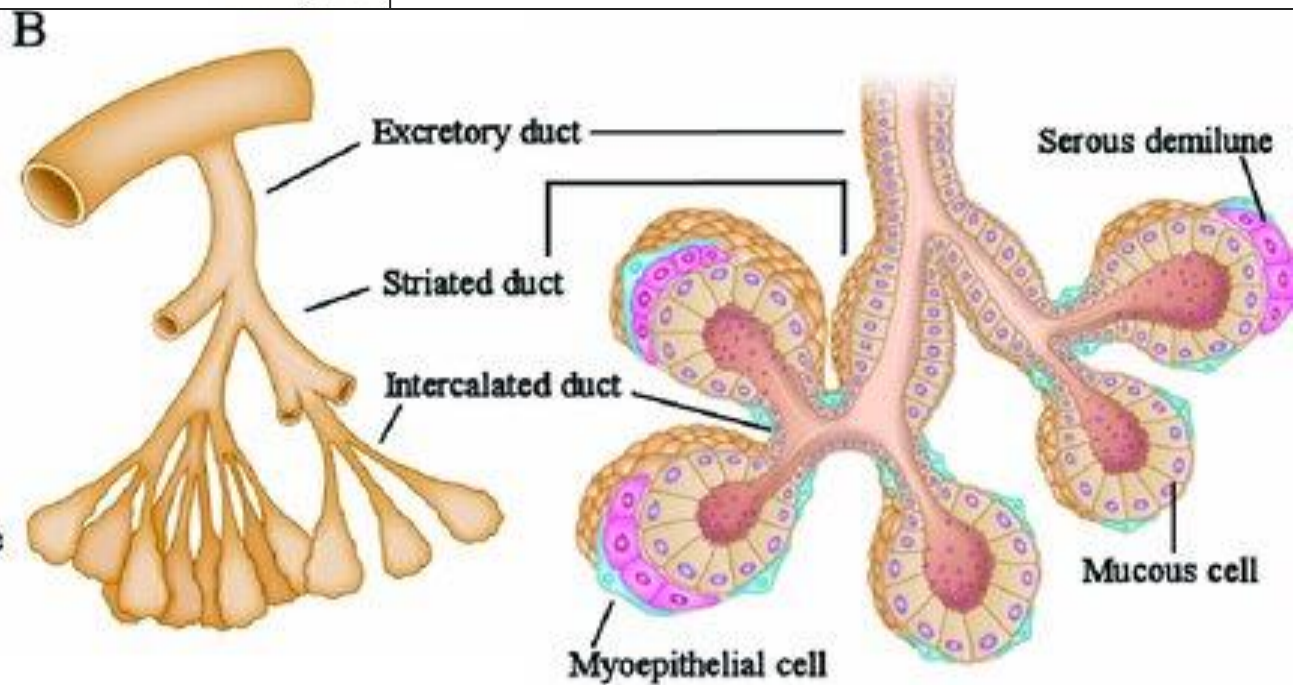
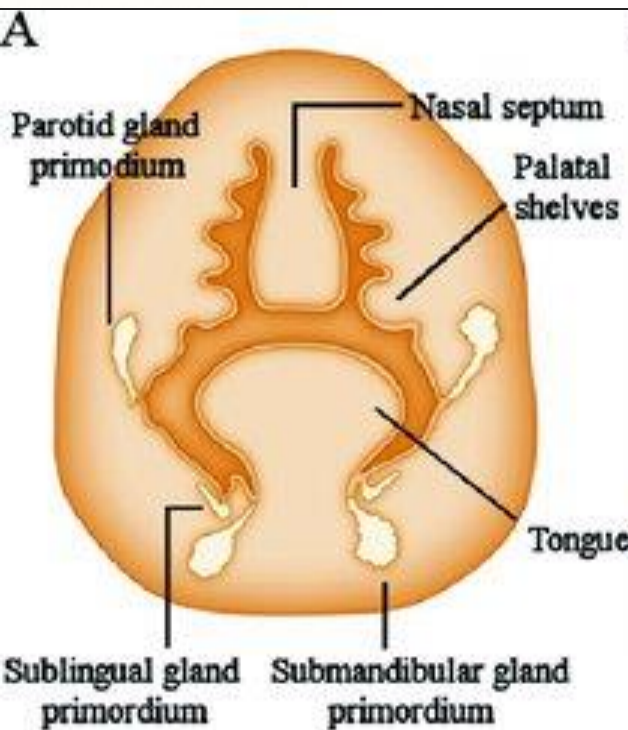
A thin **oronasal membrane** separates the two cavities.

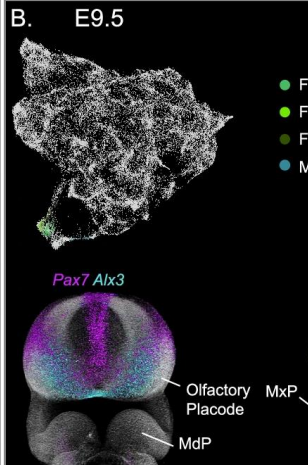
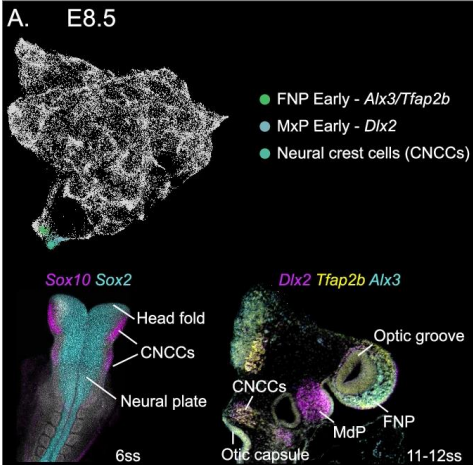
By its rupture the **primitive choanae** will form





Salivary gland development and structure. In week 6-8 of fetal life, the major and minor salivary glands develop in the oral cavity from ectodermal proliferations invading the underlying mesenchyme. (B and C) The normal salivary gland is a tubuloacinar structure composed of ductal cells, myoepithelial cells, and acinar cells.





Article

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Positional programs in early murine facial development and their role in human facial shape variability

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